Math 116: Calculus II

1 Introduction

This course is taken by the majority of students intending to major in mathematics, science, or engineering, as well as students headed for many other fields. The emphasis is on concepts and solving problems rather than on theory and proofs. Math 116 at the University of Michigan presents the concepts of calculus from four points of view: *geometric, numerical, algebraic, and verbal*. Students develop their reading, writing, and questioning skills. Topics include techniques of integration, application of these, and an introduction to sequences and series and to differential equations.

- *Course website:* `http://www.math.lsa.umich.edu/courses/116/`
- *Link to notes:* `http://www-personal.umich.edu/~chrismav/notesMath116.pdf`
- *Lecture notes based on:* Calculus: Single Variable by Hughes-Hallett, Gleason, et al., 7th Edition

Note to students

The following lecture notes are meant as a guide to the classes. They are not meant to replace the actual lessons in class. You should expect that extra material can be covered during class (especially examples and past exam questions). Nevertheless I will try to follow the notation and the overall structure of these notes as much as possible.

2 How do we measure distance traveled? (§5.1)

Visualizing distance on the velocity graph

If the velocity is positive, the total distance traveled is the area under the velocity curve.

If the velocity can be negative as well as positive, then

- **Change in position** is given by the area above axis *minus* area below the axis
- **Distance traveled** is given by the area above axis *plus* area below the axis

Left and Right sums

We denote by $v = f(t)$ any non-negative velocity function. We want to find the distance traveled between $t = a$ and $t = b$. For that, we take measurements of $f(t)$ at equally spaced times $t_0, t_1, t_2, \ldots, t_n$ with time $t_0 = a$ and $t_n = b$. Therefore, we must have that the time interval between any two consecutive measurements is

$$\Delta t = \frac{b-a}{n}, \tag{1}$$

where Δt is the change in t.

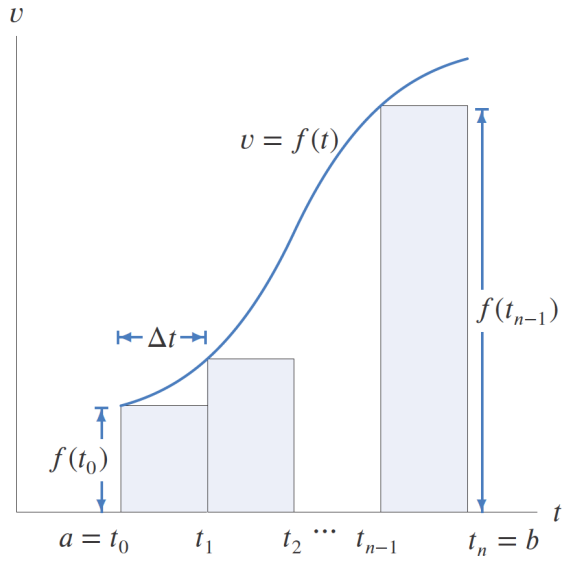


Figure 5.8: Left-hand sums

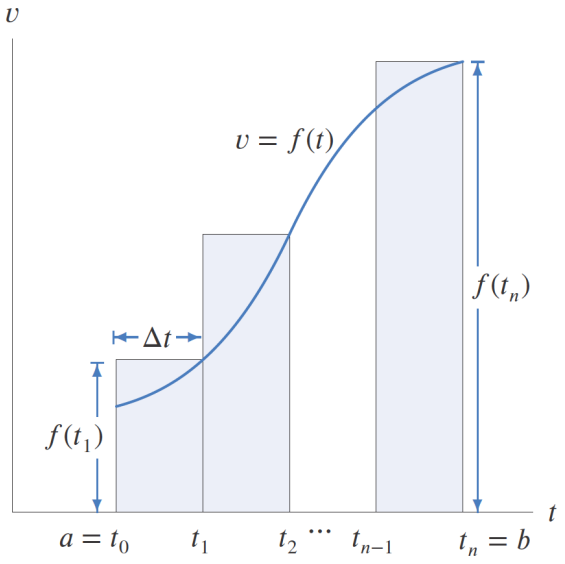


Figure 5.9: Right-hand sums

During the first time interval, so from t_0 to t_1, the velocity can be approximated by $f(t_0)$, so the distance traveled is approximately $f(t_0)\Delta t$ and during the second time interval, i.e. from t_1 to t_2, the velocity is about $f(t_1)$ and so the distance traveled is about $f(t_1)\Delta t$. Note that in the last interval the velocity is approximated by $f(t_{n-1})$ and so the last term will be $f(t_{n-1})\Delta t$.

Continuing like this and adding all the estimates, we get an estimate for the *total* distance traveled.

Left-hand sum

Total distance traveled between $t = a$ and $t = b \approx f(t_0)\Delta t + f(t_1)\Delta t + \cdots + f(t_{n-1})\Delta t$

This is called the left-hand sum because we used the value of the velocity from the left end of each time interval. It is represented by the sum of the areas of the rectangles in Figure 5.8 above.

Right-hand sum

Total distance traveled between $t = a$ and $t = b \approx f(t_1)\Delta t + f(t_2)\Delta t + \cdots + f(t_n)\Delta t$

This is called the right-hand sum because we used the value of the velocity from the right end of each time interval. It is represented by the sum of the areas of the rectangles in Figure 5.9 above.

If f is an *increasing* function, the left-hand sum is an **underestimate** and the right-hand sum is an **overestimate** of the total distance traveled.

If f is a *decreasing* function, them the roles of the left-hand sum and the right-hand sum are reversed.

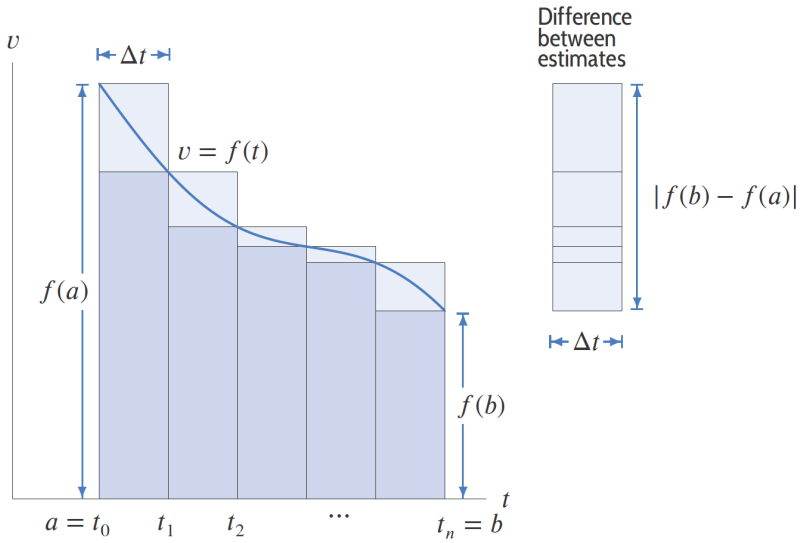


Figure 5.10: Left and right sums if f is decreasing

Note also that by making the time interval Δt between measurements small enough, we make the difference between the lower and upper estimates as small as we like.

Accuracy of estimates

For either increasing or decreasing velocity functions, the exact value of the distance traveled lies between the two estimates. Therefore, the accuracy of the estimate depends on how close these two sums are. For a function that is increasing or decreasing throughout the interval $[a, b]$, we have

$$|\text{difference between upper and lower esimates}| = |f(b) - f(a)| \cdot \Delta t. \tag{2}$$

3 The definite integral (§5.2)

Sigma notation: Recall from last time that the left and right Riemann sums. They are defined as follows

$$\text{left sum} = f(t_0)\Delta t + f(t_1)\Delta t + \cdots + f(t_{n-1})\Delta t = \sum_{i=0}^{n-1} f(t_i)\Delta t, \tag{3}$$

and

$$\text{right sum} = f(t_1)\Delta t + f(t_2)\Delta t + \cdots + f(t_n)\Delta t = \sum_{i=1}^{n} f(t_i)\Delta t. \tag{4}$$

Here the Greek capital letter Σ is used to denote a sum. The base $i = 1$ tells us for example that the sum must start at this value and the n at the top tells us that we should stop adding terms at $i = n$. The English capital letter S for sum is elongated to denote the integral symbol $\int$.

Taking the limit to get a definite integral

Let f be continuous for $a \le t \le b$. The *definite integral* of f from a to b is written in the form

$$\int_a^b f(t)\,\mathrm{d}t, \tag{5}$$

and is the limit of the left and right sums with n subdivisions for $a \le t \le b$ as n gets arbitrarily large, that is, as $\Delta t \to 0$. In other words,

$$\int_a^b f(t)\,\mathrm{d}t = \lim_{n\to\infty}(\text{left sum}) = \lim_{n\to\infty}\left(\sum_{i=0}^{n-1} f(t_i)\Delta t\right), \tag{6}$$

and similarly,

$$\int_a^b f(t)\,\mathrm{d}t = \lim_{n\to\infty}(\text{right sum}) = \lim_{n\to\infty}\left(\sum_{i=1}^{n} f(t_i)\Delta t\right). \tag{7}$$

Each of these sums is called a *Riemann sum*, f is the *integrand*, and a and b are the *limits of integration*.

Note that in computing a definite integral, as we increase the value of n which is the number of subdivisions, the over- and underestimates get closer together, and so they trap the value of the integral between them. Thus, by increasing the value of n sufficiently, we can calculate the integral to any desired accuracy.

The definite integral as an area

When $f(x) \ge 0$ (so when $f(x)$ is positive) and $a < b$, we have

$$\text{area under graph of } f \text{ and above } x\text{-axis between } a \text{ and } b = \int_a^b f(x)\,\mathrm{d}x. \tag{8}$$

As we see in the figure below, the width Δx of the rectangles goes to zero, the rectangles fit the curve of the graph more exactly, and the sum of their areas gets closer and closer to the area under the shaded curve, in Figure 5.29.

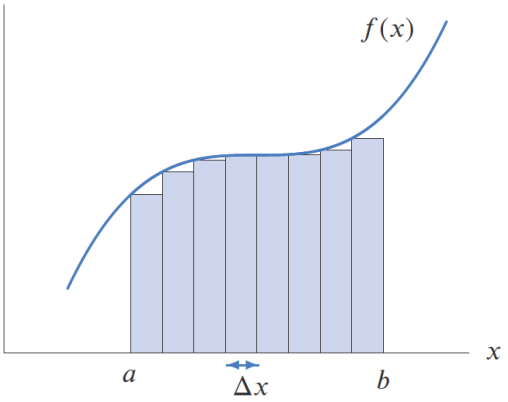


Figure 5.28: Area of rectangles approximating the area under the curve

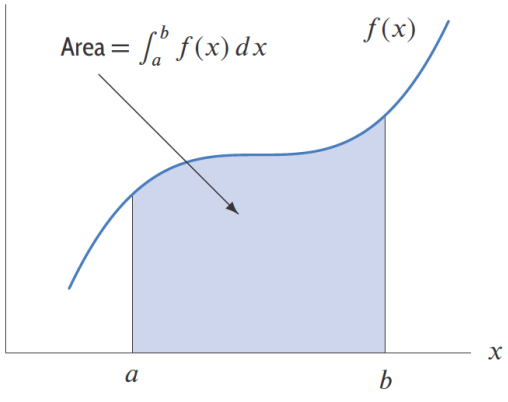


Figure 5.29: The definite integral $\int_a^b f(x)\,dx$

When $f(x)$ is not positive

If the graph lies below the x-axis, then each value of $f(x)$ is negative, so each $f(x)\Delta x$ is negative, and the area gets counted negatively. In this case, the definite integral is the negative of the area.

When $f(x)$ is positive for some x values and negative for others, and $a < b$,

$$\int_a^b f(x)\,\mathrm{d}x \tag{9}$$

is the sum of areas above the x-axis, counted positively, and areas below the x-axis counted negatively.

4 Fundamental Theorem and Interpretations (§5.3)

The fundamental theorem of calculus

Recall that the change in position can be calculated as the limit of Riemann sums of the velocity function $v = f(t)$. Thus, change in position is given by the definite integral $\int_a^b f(t)\,\mathrm{d}t$. If we let $F(t)$ denote the position function, then the change in position can be written as $F(b) - F(a)$. Thus, we have

$$\int_a^b f(t)\,\mathrm{d}t = (\text{change in position from } t = a \text{ to } t = b) = F(b) - F(a). \tag{10}$$

We also know that the position function F and velocity f are related using derivatives, through $F'(t) = f(t)$. This relation between the integral and the derivative applied to any function F with a continuous derivative $f = F'$.

The fundamental theorem of calculus

If f is continuous on the interval $[a, b]$ and $f(t) = F'(t)$, then

$$\int_a^b f(t)\,\mathrm{d}t = F(b) - F(a). \tag{11}$$

To understand the fundamental theorem of calculus, we think of $f(t) = F'(t)$ as the rate of change of the quantity $F(t)$. To calculate the total change in $F(t)$ between times $t = a$ and $t = b$, we divide the interval $a \le t \le b$ into n subintervals, each of length Δt. For each small interval, we estimate the change in $F(t)$, written as ΔF, and add these. In each subinterval we assume the rate of change of $F(t)$ is approximately constant, so that we can say

$$\Delta F \approx \text{rate of change of } F \times \text{time elapsed}. \tag{12}$$

For the subinterval, from t_0 to t_1, the rate of change $F(t)$ is approximately $F'(t_0)$, so

$$\Delta F \approx F'(t_0)\Delta t. \tag{13}$$

Similarly, for the second interval

$$\Delta F \approx F'(t_1)\Delta t. \tag{14}$$

Summing over all the subintervals, we get

$$\text{total change in } F(t) \text{ between } t=a \text{ and } t=b = \sum_{i=0}^{n-1} \Delta F \approx \sum_{i=0}^{n-1} F'(t_i)\Delta t. \tag{15}$$

We have approximated the change in $F(t)$ as a left-hand sum.

However, the total change in $F(t)$ between the times $t=a$ and $t=b$ is simply $F(b)-F(a)$. Taking the limit as $n \to \infty$ converts the Riemann sum to a definite integral and suggests the following interpretation of the Fundamental Theorem of Calculus

$$F(b)-F(a) = (\text{total change in } F(t) \text{ between } t=a \text{ and } t=b) = \int_a^b F'(t)\,\mathrm{d}t. \tag{16}$$

Thus, the definite integral of a rate of change gives the total change.

Recommended book exercises

1. From §5.1
   - Questions: 1, 11, 17–20, 23, 27, 37, 39
2. From §5.2
   - Questions: 6, 10, 21, 24, 36, 44
3. From §5.3
   - Questions: 1, 3, 5, 13, 21, 23, 27, 42

5 Theorems about definite integrals (§5.4)

Properties of limits of integration

If a, b and c are any numbers and f is a continuous function, then

- $\displaystyle\int_b^a f(x)\,\mathrm{d}x = -\int_a^b f(x)\,\mathrm{d}x.$
  The integral from b to a is the negative of the integral from a to b.
- $\displaystyle\int_a^c f(x)\,\mathrm{d}x + \int_c^b f(x)\,\mathrm{d}x = \int_a^b f(x)\,\mathrm{d}x.$
  The integral from a to c plus the integral from c to b is the integral from a to b.

Properties of sums and constant multiples of the integrand

Let f and g be continuous functions and let c be a constant.

- $\int_a^b (f(x) \pm g(x))\,\mathrm{d}x = \int_a^b f(x)\,\mathrm{d}x \pm \int_a^b g(x)\,\mathrm{d}x.$

  The integral of the sum (or difference) of two functions is the sum (or difference) of their integrals.

- $\int_a^b cf(x)\,\mathrm{d}x = c\int_a^b f(x)\,\mathrm{d}x.$

  The integral of a constant times a function is that constant times the integral of the function.

Area between curves

If the graph of $f(x)$ lies above the graph of $g(x)$ for $a \le x \le b$, then

$$(\text{area between } f \text{ and } g \text{ for } a \le x \le b) = \int_a^b (f(x) - g(x))\,\mathrm{d}x. \tag{17}$$

Using symmetry to evaluate integrals

Symmetry can be useful when it comes to evaluating definite integrals. Recall that an *even* function is symmetric about the y-axis and an *odd* function is symmetric about the origin.

Consider the following figures for example:

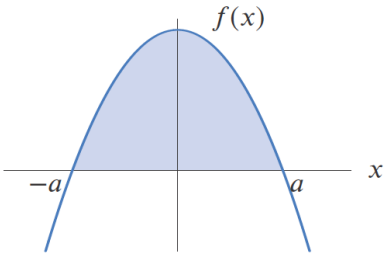


Figure 5.68: For an even function, $\int_{-a}^{a} f(x)\,dx = 2\int_0^a f(x)\,dx$

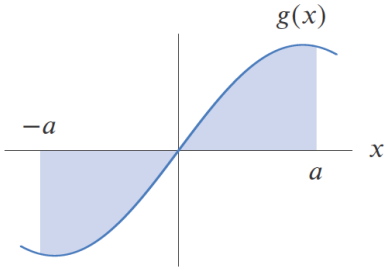


Figure 5.69: For an odd function, $\int_{-a}^{a} g(x)\,dx = 0$

It is clear that,

- If f is even, then $\int_{-a}^{a} f(x)\,\mathrm{d}x = 2\int_0^a f(x)\,\mathrm{d}x.$
- If f is odd, then $\int_{-a}^{a} f(x)\,\mathrm{d}x = 0.$

Comparison of definite integrals

Let f and g be continuous functions.

- If $m \le f(x) \le M$ for $a \le x \le b$, then $m(b-a) \le \int_a^b f(x)\,\mathrm{d}x \le M(b-a).$

- If $f(x) \le g(x)$ for $a \le x \le b$, then $\displaystyle\int_a^b f(x)\,\mathrm{d}x \le \int_a^b g(x)\,\mathrm{d}x$.

As in the textbook, you can visualize these facts from a graph as follows

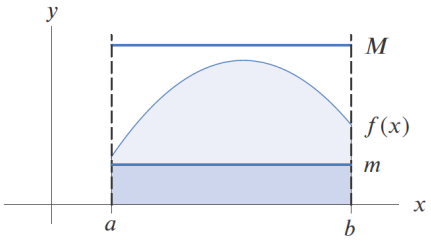


Figure 5.72: The area under the graph of f lies between the areas of the rectangles

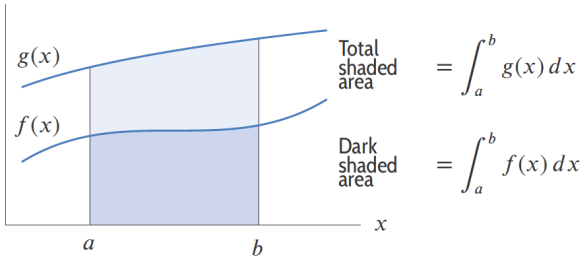


Figure 5.73: If $f(x) \le g(x)$ then $\int_a^b f(x)\,dx \le \int_a^b g(x)\,dx$

The definite integral as an average of a function

$$(\text{average value of } f \text{ from } a \text{ to } b) = \frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x. \tag{18}$$

6 Antiderivatives graphically and numerically (§6.1)

Definition: If the derivative of F is f, we call F the antiderivative of f.

Example. If v is the velocity of a car and s is the position, then $v = \mathrm{d}s/\mathrm{d}t$ and s is the antiderivative of v. As before $s + C$ is an antiderivative of v for any constant C. What does this C represent? In terms of the car, adding C to s is equivalent to adding C to the odometer reading. Adding a constant to the odometer reading simply means measuring distance from a different starting point, which does not change the car's velocity.

Sketch f from f'

We are looking for the graph of f whose slope at any point is equal to the value of f' there.

- If f' is above the x-axis, f is increasing.
- If f' is below the x-axis, f is decreasing.
- If f' is increasing, f is concave up.
- If f' is decreasing, f is concave down.

Computing antiderivatives using definite integrals

A graph of f' shows where f is increasing and where decreasing. We can calculate the actual value of the function f using the Fundamental Theorem of Calculus. If f' is continuous, then

$$\int_a^b f'(x)\,\mathrm{d}x = f(b) - f(a). \tag{19}$$

If we know $f(a)$, we can estimate $f(b)$ by computing the definite integral using area or Riemann sums.

7 Constructing antiderivatives analytically (§6.2)

The indefinite integral

All antiderivatives of f are of the form $F(x) + C$. We introduce a notation for the general antiderivative that looks like the definite integral **without** the limits of integration and it is called the *indefinite integral*. It is written

$$\int f(x)\,\mathrm{d}x = F(x) + C. \tag{20}$$

Note that the definite integral $\int_a^b f(x)\,\mathrm{d}x$ is a number, whereas the indefinite integral $f(x)\,\mathrm{d}x$ is a family of functions.

Constructing antiderivatives analytically

- If $F'(x) = 0$ on an interval, then $F(x) = C$ on this interval, where C is some constant.
- If F and G are both antiderivatives of f on an interval, then $G(x) = F(x) + C$.
- If k is a constant, $\int k\,\mathrm{d}x = kx + C$.
- $\int x^n\,\mathrm{d}x = \dfrac{x^{n+1}}{n+1} + C, n \neq 1$

  Remember it as follows: add one to the power and divide by the new power!
- $\int \dfrac{1}{x}\,\mathrm{d}x = \ln|x| + C.$
- $\int e^x\,\mathrm{d}x = e^x + C.$
- $\int \cos(x)\,\mathrm{d}x = \sin(x) + C.$
- $\int \sin(x)\,\mathrm{d}x = -\cos(x) + C.$ Mnemonic: **s**(in) **i**(ntegrates) **n**(egatively)

Properties of antiderivatives: sums and constant multiples

In indefinite integral notation

- $\int (f(x) \pm g(x))\,\mathrm{d}x = \int f(x)\,\mathrm{d}x \pm \int g(x)\,\mathrm{d}x.$

  An antiderivative of the sum/difference of two functions is the sum/difference of their antiderivatives.
- $\int cf(x)\,\mathrm{d}x = c\int f(x)\,\mathrm{d}x.$

  An antiderivative of a constant times a function is the constant times an antiderivative of the function.

Using antiderivatives to compute definite integrals

We can employ the Fundamental theorem of calculus to compute definite integrals as follows. If we denote $F(b) - F(a)$ by $F(x)|_a^b$, and we know that $F' = f$, with f continuous, then

$$\int_a^b f(x)\,\mathrm{d}x = F(x)|_a^b = F(b) - F(a). \tag{21}$$

Recommended book exercises

1. From §5.4
   - Questions: 1–6, 11, 34, 38, 40
2. From §6.1
   - Questions: 3, 8, 10, 22, 25, 37, 39
3. From §6.2
   - Questions: 57–63 (odd), 76, 82, 86

8 Second Fundamental Theorem of Calculus (§6.4)

Construction theorem for antiderivatives: If f is a continuous function on an interval, and if a is any number in that interval, then the function F defined on the interval as follows is an antiderivative of f:

$$F(x) = \int_a^x f(t)\,\mathrm{d}t. \tag{22}$$

Examples. Find the following derivatives

- $\dfrac{\mathrm{d}}{\mathrm{d}x}\displaystyle\int_0^x \cos(t^2)\,\mathrm{d}t$
- $\dfrac{\mathrm{d}}{\mathrm{d}t}\displaystyle\int_4^t \sin(\sqrt{x})\,\mathrm{d}x$
- $\dfrac{\mathrm{d}}{\mathrm{d}x}\displaystyle\int_1^x (1+t)^{200}\,\mathrm{d}t$
- $\dfrac{\mathrm{d}}{\mathrm{d}x}\displaystyle\int_2^x \ln(t^2+1)\,\mathrm{d}t$

Using the chain rule, calculate the derivatives

- $\dfrac{\mathrm{d}}{\mathrm{d}x}\displaystyle\int_0^{x^2} \ln(t^2+1)\,\mathrm{d}t$
- $\dfrac{\mathrm{d}}{\mathrm{d}t}\displaystyle\int_1^{\sin t} \cos(x^2)\,\mathrm{d}x$

- $\dfrac{\mathrm{d}}{\mathrm{d}t}\displaystyle\int_{2t}^{4}\sin(\sqrt{x})\,\mathrm{d}x$
- $\dfrac{\mathrm{d}}{\mathrm{d}x}\displaystyle\int_{-x^2}^{x^2}\exp(t^2)\,\mathrm{d}t$

Recommended book exercises

1. From §6.4
    - Questions: 5, 7, 9, 11, 17, 27, 30–33, 34, 36, 38–41

9 Integration by substitution (§7.1)

Integration by recognition

Sometimes a good strategy is to find simple antiderivatives is by *guessing* or "recognizing" the answer. This is done using your knowledge of differentiation rules. You can then check your answer by differentiating it. If the result from differentiating your answer gives you the integrand then you are done.

Recall the **chain rule**: $\dfrac{\mathrm{d}}{\mathrm{d}x}(f(g(x))) = f'(g(x))g'(x)$. So if a function has this form then its antiderivative must be $f(g(x))$.

Example. Find $\displaystyle\int 5x^4\sin(x^5)\,\mathrm{d}x$.

Looking at the integrand, we see that the function $5x^4\sin(x^5)$ is the result of applying the chain rule on $-\cos(x^5)$. Equivalently,

$$\frac{\mathrm{d}}{\mathrm{d}x}(-\cos(x^5)) = 5x^4\sin(x^5).$$

Since this is our integrand, we know that

$$\int 5x^4\sin(x^5)\,\mathrm{d}x = -\cos(x^5) + C,$$

where C is called the *constant of integration*.

The method of substitution

It is often the case that the integrand is a complicated expression, and so it is helpful to formalize our method of recognition as follows

Making a substitution: Let w be the *inside function* and $\mathrm{d}w = w'(x)\,\mathrm{d}x = \frac{\mathrm{d}w}{\mathrm{d}x}\,\mathrm{d}x$.

Same example. Find $\displaystyle\int 5x^4\sin(x^5)\,\mathrm{d}x$.

The inside function in this case is x^5, so let $w = x^5$. This implies that $\mathrm{d}w = 5x^4\,\mathrm{d}x$. Now let us rewrite the integral in terms of the new variable w. It becomes

$$\int 5x^4\sin(x^5)\,\mathrm{d}x = \int \sin(w)\,\mathrm{d}w.$$

This is now easy to integrate. The result is $\int \sin(w)\,\mathrm{d}w = -\cos(w) + C$. Finally we need to substitute back for the original variable x. Therefore, the final answer must be

$$\int 5x^4 \sin(x^5)\,\mathrm{d}x = -\cos(x^5) + C.$$

Attention! Substitution works if the integrand contains the derivative of the inside function, to within a constant factor. If this factor is not *constant* then we cannot use this method.

To use substitution to find definite integrals

Either

- Compute the indefinite integral, expressing an antiderivative in terms of the original variable, and then evaluate the result at the original limits,

or

- convert the original limits to new limits in terms of the new variable and do not convert the antiderivative back to the original variable.

Recommended book exercises

1. From §7.1
   - Questions: 84, 88–94, 99, 100, 102, 112, 114, 117, 129, 147, 157, 159

10 Integration by parts (§7.2)

The method of substitution reverses the chain rule. In this section, we introduce *integration by parts*, which is a method based on the *product rule*.

Recall that

$$\frac{\mathrm{d}(uv)}{\mathrm{d}x} = v\frac{\mathrm{d}u}{\mathrm{d}x} + u\frac{\mathrm{d}v}{\mathrm{d}x}.$$

Rearranging gives

$$u\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{\mathrm{d}(uv)}{\mathrm{d}x} - v\frac{\mathrm{d}u}{\mathrm{d}x}.$$

Now, integrating each term with respect to x gives

$$\int u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = \int \frac{\mathrm{d}(uv)}{\mathrm{d}x}\,\mathrm{d}x - \int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x.$$

Since differentiating a function and then integrating leaves the function unchanged, we can simplify $\displaystyle\int \frac{\mathrm{d}(uv)}{\mathrm{d}x}\,\mathrm{d}x$ to uv, and this gives the **integration by parts** formula:

$$\int u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = uv - \int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x. \tag{23}$$

This formula enables us to exchange an integral which is complicated $\int u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x$ for a simpler one $\int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$.

How to choose u and $\frac{\mathrm{d}v}{\mathrm{d}x} = v'$:

- Choose v' so that it is easy to find v.
- It is helpful if u' is simpler than u.
- It is helpful if v is simpler than v'.

In general, we will *usually* let $u =$ any terms of the form x^n, but there is one exception to this and that is when there is an $\ln(\cdot)$ term. In that case, we let $u = \ln(\cdot)$.

Notes. Integration by parts involves integrating in two separate stages (first the uv term then the $\int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$). Any limits can be applied separately to each other.

Sometimes you need to do integration by parts more than once.

Example. Find $\int x\cos(x)\,\mathrm{d}x$.

Let $u = x$ and $\frac{\mathrm{d}u}{\mathrm{d}x} = \cos(x)$.

$$u = x \qquad \Rightarrow \qquad \frac{\mathrm{d}u}{\mathrm{d}x} = 1 \tag{24}$$

$$\frac{\mathrm{d}v}{\mathrm{d}x} = \cos(x) \qquad \Rightarrow \qquad v = \sin(x) \tag{25}$$

Using the integration by parts formula:

$$\begin{aligned}\int x\cos(x)\,\mathrm{d}x &= x\sin(x) - \int \sin(x)\times 1\,\mathrm{d}x \\ &= x\sin(x) + \cos(x) + C.\end{aligned}$$

Another example. Find $\int x^2\ln(x)\,\mathrm{d}x$.

Recommended book exercises

1. From §7.2
   - Questions: 55, 59, 61, 70, 76, 81

11 Partial fractions with two distinct linear factors (§7.4)

We can use **partial fractions** to integrate expressions.

Recall that you can add (or subtract) two or more simple fractions as long as their denominators are the same.

You can split a fraction with two linear factors in the denominator into partial fractions. For example

$$\frac{x-1}{(x+1)(x+3)} = \frac{2}{x+3} - \frac{1}{x+1}.$$

In general, an expression with two linear terms in the denominator such as $\dfrac{11}{(x-3)(x+2)}$ can be split into two partial fractions of the form $\dfrac{A}{(x-3)} + \dfrac{B}{(x+2)}$, where A and B are constants to be determined.

There are two methods for achieving this: (a) by substitution and (b) by equating coefficients. Here we demonstrate the method by substitution. The method of equating coefficients will be demonstrated in class.

Example method. Split $\dfrac{6x-2}{(x-3)(x+1)}$ into partial fractions.

1. $\dfrac{6x-2}{(x-3)(x+1)} = \dfrac{A}{(x-3)} + \dfrac{B}{(x+1)}$

2. Add the two fractions to obtain: $\dfrac{6x-2}{(x-3)(x+1)} = \dfrac{A(x+1)+B(x-3)}{(x-3)(x+1)}$.

3. Since this is an equivalence relation we set the numerators equal to each other:

$$6x-2 = A(x+1)+B(x-3).$$

4. To find A we want to set the contribution from B to zero, so substitute $x=3$. This gives

$$\begin{aligned} 6(3)-2 &= A(3+1)+B(3-3) \\ 16 &= 4A \\ A &= 4. \end{aligned}$$

5. To find B substitute $x=-1$:

$$\begin{aligned} 6(-1)-2 &= A(-1+1)+B(-1-3) \\ -8 &= -4B \\ B &= 2. \end{aligned}$$

6. Substitute back into step 1:

$$\frac{6x-2}{(x-3)(x+1)} = \frac{4}{(x-3)} + \frac{2}{(x+1)}.$$

Note. If there is a quadratic in the denominator which cannot be factored, we need an expression of the form $Ax + B$ in the numerator.

Exercise. Find $\displaystyle\int \frac{2x^2 - x - 1}{(x^2+1)(x-2)}\,\mathrm{d}x$.

Hint: Use partial fractions of the form $\dfrac{Ax+B}{x^2+1}$ and $\dfrac{C}{x-2}$.

Recommended book exercises

1. From §7.4
    - Questions: 25–26, 38, 46, 66, 82

12 Numerical methods for definite integrals

The midpoint rule

Recall that for left and right Riemann sums, we were using the left- and right-hand endpoints of the subintervals as the heights of the rectangles. For the **midpoint rule**, we will use the midpoint of each of the subintervals as shown in figure 7.5.

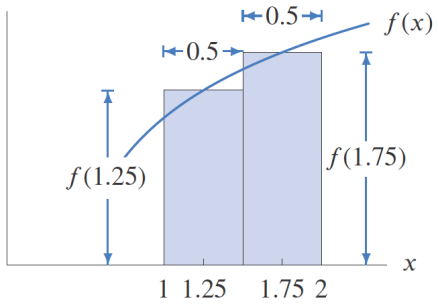


Figure 7.5: Midpoint rule with two subdivisions

Note that often, evaluating f at the midpoint of each subdivisions gives a better approximation to the area under the curve than evaluating f at either end.

Thus, we have three ways of estimating an integral using a Riemann sum

1. The *left rule* uses the left endpoint of each subinterval
2. The *right rule* uses the right endpoint of each subinterval.
3. The *midpoint rule* uses the midpoint of each subinterval.

We write LEFT(n), RIGHT(n), and MID(n) to denote the results obtained by using these rules with n subdivisions.

The trapezoid rule

We have just seen how the midpoint rule can have the effect of balancing out the errors of the left and right rules. another way of balancing these errors is to average the results from the left and right rules. This approximation is called the **trapezoid rule:**

$$\text{TRAP}(n) = \frac{\text{LEFT}(n) + \text{RIGHT}(n)}{2}. \tag{26}$$

The trapezoid rule averages the values of f at the left and right endpoints of each subinterval and multiplies by Δx. This is the same as approximating the area under the graph of f in each subinterval by a trapezoid, as shown in figure 7.7.

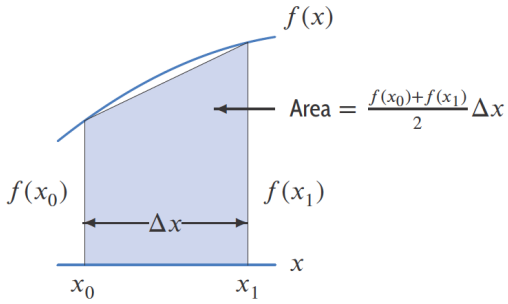


Figure 7.7: Area used in the trapezoid rule

Is the approximation an over- or underestimate?

The trapezoid rule

- If the graph of the function is *concave down* on $[a, b]$, then each trapezoid lies below the graph and the trapezoid rule gives an *underestimate*.
- If the graph of the function is *concave up* on $[a, b]$, then each trapezoid lies above the graph and the trapezoid rule gives an *overestimate*.



Figure 7.8: Error in the trapezoid rule

The midpoint rule

To understand the relationship between the midpoint rule and concavity, take a rectangle whose top intersects the curve at the midpoint of a subinterval. Draw a tangent to the curve at the midpoint and this will give the trapezoid. The midpoint rectangle and the new trapezoid have the same area, because the shaded triangles are congruent. Hence, if the graph of the function is concave down, the midpoint rule overestimates, and if the graph is concave up, the midpoint rule underestimates.

If the graph of f is concave down on $[a, b]$, then

$$\text{TRAP}(n) \leq \int_a^b f(x)\,dx \leq \text{MID}(n). \tag{27}$$

If the graph of f is concave up on $[a, b]$, then

$$\text{MID}(n) \leq \int_a^b f(x)\,dx \leq \text{TRAP}(n). \tag{28}$$

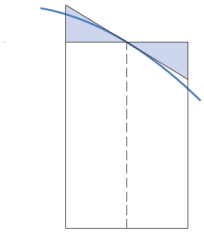


Figure 7.9: Midpoint rectangle and trapezoid with same area

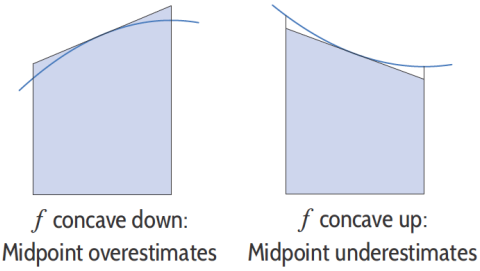


Figure 7.10: Error in the midpoint rule

The error is defined as

$$\text{error} = \text{actual value} - \text{approximate value}.$$

In general, the midpoint rule and the trapezoid rule are more accurate than the left and right Riemann sums.

1. From §7.5
    - Questions: 1, 3, 5, 21, 22, 34